

HELIX

Closed-Loop Autonomous Discovery OS

Technical Master Specification & Enterprise Presentation Dossier

Purpose

A presentation-ready consolidation of the supplied HELIX specification: market problem, architecture, data models, autonomous discovery pipeline, mathematical foundations, industry cases, platform guide, enterprise positioning, objection handling, and a recommended presentation structure.

DOCUMENT TYPE	VERSION	SOURCE BASIS
Technical + Business Dossier	Presentation-ready	Supplied HELIX master specification

Note: quantitative market, performance, ROI, and case-study figures in this dossier are reproduced from the supplied source material and have not been independently verified in this conversion.

Contents

#	Section
1	Executive Summary
2	Market Need & Product Philosophy
3	HELIX System Architecture
4	Core Data Models
5	Eight-Stage Closed-Loop Pipeline
6	Mathematical & Algorithmic Foundations
7	Industry Case Studies & ROI
8	Control Room & Platform User Guide
9	Enterprise Sales Playbook
10	Presentation-Ready Slide Structure
11	Presenter Talking Points
12	Source Notes & Claims Handling

1. Executive Summary

HELIX is positioned in the supplied specification as an autonomous closed-loop operating system for scientific and industrial R&D. Instead of stopping at recommendations, it connects hypothesis generation, experimental design, risk controls, physical or simulated execution, measurement, Bayesian updating, knowledge-graph updates, and cryptographic provenance.

The core problem

- R&D organizations generate large volumes of research insight, but digital reasoning, physical execution, instrumentation, and organizational memory are often disconnected.
- The source frames three bottlenecks: the open-loop disconnect, reproducibility/decomposition problems, and the dead-end branch trap.

The HELIX response

- Athena generates hypotheses and priors.
- Daedalus designs experiments and controls.
- Pythia evaluates risk and statistical outcomes.
- Hermes orchestrates queues, hardware, and telemetry.
- Mnemosyne maintains the evolving knowledge graph.
- A scientist-in-the-loop gate constrains higher-risk actions.
- A cryptographic provenance ledger records actors, actions, cycles, artifacts, and hashes.

Central thesis: the system's proposed moat is the closed loop itself—the continuous connection between reasoning, experimentation, evidence, and institutional memory.

What this document contains

- A technical explanation suitable for engineering and architecture discussions.
- A business explanation suitable for CSO, automation, IP/compliance, and CTO audiences.
- A presentation structure that can be converted directly into slides.
- Speaker notes and a claims-handling section so quantitative statements are presented with appropriate source attribution.

2. Market Need & Product Philosophy

2.1 R&D productivity bottleneck

The supplied specification states that global enterprise R&D expenditure exceeds **\$2.5 trillion annually** and argues that research productivity has declined across pharmaceuticals, materials science, energy storage, and advanced manufacturing. The source attributes this to three recurring bottlenecks.

BOTTLENECK	WHAT IT MEANS	HELIX RESPONSE
Open-loop disconnect	Digital recommendations do not automatically become physical experiments, and experimental results do not automatically feed the next decision.	Connect AI agents to robotics, testbeds, sensors, and digital twins.
Reproducibility / context loss	Implicit experimental conditions, environmental noise, and instrument drift can be difficult to capture and reproduce.	Capture structured protocols, telemetry, outcomes, and provenance.
Dead-end branch trap	Failed or weak hypotheses can consume significant time when negative evidence is not systematically retained and reused.	Require falsifiers and retire failed mechanism branches into the knowledge graph.

2.2 Why a chatbot alone is not the proposed solution

- Physical feedback: a language model does not itself observe empirical consequences.
- Falsification rigor: the system requires explicit kill criteria rather than only plausible narrative output.
- Compounding memory: HELIX proposes a live enterprise ontology that changes as experiments produce evidence.

Product distinction: HELIX is described as an operating layer around an experimental loop, rather than as a standalone conversational assistant.

2.3 Closed-loop concept

[Hypothesis] → [Experiment Design] → [Risk Gate] → [Execution] → [Measurement]
↑ ↓
■■■■■■■■ [Bayesian Update + Knowledge Graph + Provenance] ■■■■■■

3. HELIX System Architecture

3.1 Layered architecture

LAYER	COMPONENTS	ROLE
Control Room / API	React 19 / Vite UI; State Store; Engine Dispatcher	Campaign control, monitoring, API access, human interaction.
Multi-Agent AI Core	Athena; Daedalus; Pythia; Hermes; Mnemosyne	Hypothesis, experiment design, evaluation, orchestration, and organizational memory.
Execution & Hardware	Robotic liquid handler / cell foundry; manufacturing testbed / LPBF printer; edge sensors; digital twin	Physical execution, simulation, sensing, and telemetry.
Security & Governance	SHA-256 provenance ledger; scientist-in-the-loop gate	Auditability and policy-bounded autonomy.

3.2 Agent responsibilities

AGENT	RESPONSIBILITY	OUTPUT / CONTROL
Athena	Hypothesis & priors engine	Hypotheses, prior probabilities, uncertainty, expected lift, falsifiers.
Daedalus	Protocol & DOE architect	Experiment protocol, controls, hardware target, duration, cost, risk.
Pythia	Risk gatekeeper & evaluator	Risk decision, outcome classification, confidence interval, readout.
Hermes	Telemetry & queue orchestrator	Hardware scheduling, queueing, execution orchestration, telemetry flow.
Mnemosyne	Knowledge graph engine	Ontology delta, updated graph state, compounding organizational memory.

3.3 End-to-end flow

```
UI → State Store → Athena → Daedalus → Pythia
↓ ↓
High Risk Low/Approved
↓ ↓
Human Gate Hermes
■■■■■■■■→ Execution Layer
↓ telemetry
Pythia
↓
Mnemosyne
↓
Provenance Ledger
↓
UI
```

4. Core Data Models

The supplied TypeScript schemas define the minimum structured objects required to make a campaign traceable from research question through experiment and evidence update.

Campaign — Master discovery-loop object

- Identity: id, name, domain, owner, question
- Lifecycle: stage, cycle, running, waitingApproval, autoApproveLow
- State: hypotheses, selected hypothesis, experiment, last result
- Knowledge: nodes, edges, provenance, history
- Metrics: knowledge gain, model calibration, cycles closed, experiments queued

Hypothesis — Athena's Bayesian prior unit

- Statement and mechanism
- Prior probability and epistemic uncertainty
- Expected lift
- Mandatory falsifier / kill criterion
- Evidence, agents, and optional human attribution

Experiment — Daedalus's execution protocol

- Linked hypothesis and protocol
- Hardware type and site
- Duration, cost, and risk
- Controls and readout
- Optional time-series telemetry

Result — Pythia's statistical readout

- Outcome: supports / partial / falsifies
- Metric, value, delta
- 95% confidence interval
- Interpretive notes

GraphNode / GraphEdge — Mnemosyne's knowledge graph

- Nodes represent entities, properties, mechanisms, constraints, and measurements
- Node confidence and cycle metadata
- Edges encode relations such as exhibits, catalyzes, contradicted-by, and limits

Provenance — Cryptographic audit entry

- Timestamp, cycle, actor, actor kind, action, artifact
- SHA-256 hash digest

5. Eight-Stage Closed-Loop Pipeline

1 — INGEST

Aggregate literature, prior posteriors, and sensor baselines.

2 — HYPOTHESIZE

Athena generates hypotheses with priors and mandatory kill criteria.

3 — DESIGN

Daedalus formulates protocol, controls, and risk estimate.

4 — APPROVE

High-risk work, and medium-risk work when auto-approval is disabled, pauses for scientist signature.

5 — QUEUE

Hermes schedules the job on the target robotic lab, testbed, or other execution environment.

6 — RUN

Physical execution or digital-twin simulation streams sensor telemetry.

7 — MEASURE

Pythia classifies the outcome and computes statistical readouts including confidence intervals.

8 — UPDATE

Mnemosyne updates the ontology, posterior beliefs, cycle count, and cryptographic audit trail.

Closed-loop invariant: every cycle should produce an explicit evidence update and a traceable change to the system's knowledge state.

Approval logic

```
IF risk = HIGH
→ mandatory scientist signature
ELSE IF risk = MEDIUM AND autoApproveLow = false
→ mandatory scientist signature
ELSE
→ auto-clear to queue
```

6. Mathematical & Algorithmic Foundations

6.1 Bayesian posterior update

The source specifies Bayes' theorem for updating the belief in a hypothesis after an experiment:

$$P(H \mid E) = P(E \mid H) \cdot P(H) / P(E)$$

- A supporting outcome increases the hypothesis probability toward a stated upper bound.
- A falsifying outcome triggers the hypothesis kill criterion, retires the mechanism branch, and sets the posterior to zero in the source specification.

6.2 Epistemic uncertainty reduction

The source models uncertainty as decreasing with confidence while respecting a floor:

$$\sigma = \max(\sigma_{\text{floor}}, \sigma - \alpha \cdot \text{Confidence} + \epsilon)$$

6.3 Provenance hashing

Each provenance record is chained to the previous record through a SHA-256 digest:

$$\text{Hash} = \text{SHA256}(\text{Actor} \parallel \text{Action} \parallel \text{Cycle} \parallel \text{Timestamp} \parallel \text{Hash}_{\text{prev}})$$

This creates a tamper-evident sequence in which later entries depend on earlier hashes.

7. Industry Case Studies & ROI

The following case studies reproduce the quantitative claims supplied in the source. They should be presented as HELIX source claims unless independently validated before external publication.

Energy Storage — Li-Metal Solid Electrolytes

- Problem: lithium dendrite penetration and low room-temperature ionic conductivity.
- Approach: explore sulfide-halide composites, cold-press compaction densities, and space-charge layers.
- Source-reported impact: 12.4 mS/cm ionic conductivity in 11 days, versus 14 months of traditional bench chemistry.

Drug Discovery — JAK2 Allosteric Inhibitors

- Problem: off-target JAK1 binding and hERG channel binding.
- Approach: C-helix glycine swing hypothesis, blinded SPR assays, and high-risk scientist sign-off.
- Source-reported impact: >42× selectivity over JAK1 with 0 hERG liability in 6 closed-loop cycles.

Advanced Manufacturing — Ti-6Al-4V LPBF

- Problem: keyhole gas voids and excessive thermal residual stress.
- Approach: queued skywriting-delay protocols with real-time thermal-camera telemetry.
- Source-reported impact: keyhole porosity below 0.14% while residual stress remained under 165 MPa.

Source-supplied vertical snapshot

VERTICAL	SOURCE-SUPPLIED CLAIM
Battery science	14× faster discovery target series
Pharmacology	\$12M saved per target series
Additive manufacturing	99.8% part density achieved
Agriculture	18% NUE lift without protein loss

8. Control Room & Platform User Guide

8.1 Selecting campaigns

- Use the campaign navigation bar to switch between active research loops.

8.2 Loop controls

- Run loop resumes continuous cycling.
- Pause loop halts execution.
- Auto-clear low risk controls whether low-risk work can proceed without a scientist signature.

8.3 Manual hypothesis override

- Select a hypothesis on Athena's board to make it the active protocol candidate for Daedalus.

8.4 Scientist approval gate

- Review the safety alert and protocol details when Pythia halts a run.
- Approve & queue or reject and re-hypothesize.

8.5 Hardware & telemetry

- Monitor instrument status, site location, and queue position.
- Inspect live telemetry such as impedance, temperature, flow rate, and peak intensity.

8.6 Convergence analytics

- Track Knowledge Gain % across cycle history.
- Review Model vs. Reality Calibration to compare surrogate predictions with empirical results.

8.7 Audit ledger & export

- Search by hash digest, actor, or cycle.
- Export a signed CSV audit artifact for compliance workflows.

8.8 Campaign and hypothesis creation

- Create a campaign with title, domain, scientific question, and lead scientist.
- Propose a hypothesis with title, statement, causal mechanism, falsifier, and scientist attribution.

9. Enterprise Sales Playbook

9.1 Buyer personas

BUYER	PAIN POINT	HELIX VALUE PROPOSITION
Chief Scientific Officer	Long R&D timelines; repeated failed experiments.	Compounding knowledge graph and explicit falsifiers.
VP Laboratory Automation	Expensive robotics waiting for manual scripts.	Continuous queue orchestration and 24/7 hardware utilization.
Head of IP & Regulatory Compliance	Difficulty proving priority dates and audit trails.	Cryptographic provenance and signed CSV audit exports.
CTO / Head of AI	LLM point solutions without physical verification.	Closed-loop active learning with safety gates.

9.2 Objection handling

Question: LIMS / ELN overlap

Source-positioned answer: The source positions LIMS and ELNs as systems that store information, while HELIX adds an active discovery loop over those systems.

Question: Safety and expensive experiments

Source-positioned answer: Risk is explicitly classified and higher-risk protocols are held for scientist approval.

Question: Confidentiality

Source-positioned answer: The source proposes air-gapped deployment using private cloud or on-premise infrastructure with local models and graph databases.

Question: Noisy sensor data

Source-positioned answer: Pythia is described as using Bayesian noise modeling and confidence bounds; unusual observations can lead to a partial outcome and a follow-up control trial.

Question: ROI timing

Source-positioned answer: The source claims initial ROI can come from eliminating dead-end research branches and states a 30–60 day initial window; this is a source claim that should be validated before commercial use.

10. Presentation-Ready Slide Structure

The following 14-slide sequence is designed to explain HELIX to a mixed technical + executive audience without overwhelming the audience with implementation detail too early.

SLIDE	CONTENT	RECOMMENDED VISUAL
1. Title — HELIX	One-line thesis: Closed-loop autonomous discovery for enterprise R&D.	Visual: simple loop graphic.
2. The R&D Productivity Problem	Show the three bottlenecks: open-loop disconnect, reproducibility/context loss, dead-end branches.	Visual: three-column bottleneck diagram.
3. Why Chatbots Alone Stop Short	Contrast language generation with physical feedback, falsification, and live enterprise memory.	Visual: chatbot vs. closed-loop system.
4. HELIX in One Diagram	Show the full flow from hypothesis to experiment to evidence to knowledge update.	Visual: end-to-end architecture.
5. The Five AI Agents	Athena, Daedalus, Pythia, Hermes, Mnemosyne.	Visual: agent responsibility map.
6. Safety & Governance	Explain risk classes, scientist-in-the-loop gate, and auditability.	Visual: approval decision tree.
7. Eight-Stage Discovery Cycle	Walk through ingest → hypothesize → design → approve → queue → run → measure → update.	Visual: numbered pipeline.
8. Data & Knowledge Model	Campaign, hypothesis, experiment, result, graph, provenance.	Visual: entity relationship view.
9. Bayesian Learning	Show posterior update, uncertainty reduction, and evidence accumulation.	Visual: hypothesis confidence over cycles.
10. Hardware & Digital Twins	Explain robotics, testbeds, sensors, and simulation as execution targets.	Visual: execution layer.
11. Industry Cases	Battery, drug discovery, additive manufacturing; clearly label quantitative values as source claims.	Visual: three vertical cards.
12. Enterprise Buyer Value	Map CSO, lab automation, compliance, and CTO concerns to HELIX capabilities.	Visual: persona-to-value matrix.
13. Objections & Deployment	LIMS/ELN, safety, confidentiality, noisy data, ROI.	Visual: objection → architecture response.
14. Closing Thesis	The source's closing idea: the differentiating asset is the loop connecting models, experiments, evidence, and memory.	Visual: closed loop with compounding knowledge.

11. Presenter Talking Points

Opening — 30 seconds

“HELIX is not positioned as another chatbot for researchers. It is a closed-loop operating system that connects hypotheses to experiments, experiments to evidence, and evidence back into institutional memory.”

Problem — 60 seconds

“The central issue is not simply generating more ideas. It is converting ideas into controlled experiments, learning from negative results, and ensuring every cycle improves the next decision.”

Architecture — 90 seconds

“Five specialized agents divide the work: Athena reasons about hypotheses, Daedalus designs experiments, Pythia evaluates risk and evidence, Hermes orchestrates execution, and Mnemosyne preserves the evolving knowledge graph.”

Safety — 45 seconds

“Autonomy is bounded. High-risk and configured medium-risk protocols can stop at a scientist gate before hardware execution.”

Business value — 60 seconds

“The business proposition is compounding research throughput: less manual coordination, fewer repeated dead ends, better utilization of automation, and a structured memory of what has and has not worked.”

Closing — 30 seconds

“The system’s proposed moat is not a single model. It is the loop: reasoning, execution, measurement, learning, and provenance operating as one system.”

12. Source Notes & Claims Handling

12.1 Basis of this document

This dossier was converted from the supplied HELIX Technical Master Specification & Enterprise Sales Playbook. The structure, terminology, architecture, agent names, data models, pipeline stages, mathematical formulations, case-study figures, user workflow, buyer personas, and objection responses are based on that source.

12.2 Quantitative claims

- The supplied source contains quantitative claims about global R&D expenditure, reproducibility, cycle performance, selectivity, hERG liability, porosity, residual stress, agricultural NUE, savings, and ROI timing.
- Those values have been preserved for presentation completeness but are not independently verified in this conversion.
- For an investor, customer, regulatory, or scientific presentation, attach the original studies, internal validation reports, experiment logs, or third-party evidence behind each material claim.

12.3 Technical consistency check before external publication

- Confirm the provenance-hashing terminology: the source describes an “append-only murmur hash digest” but provides a SHA-256 chaining formula. Use one precise implementation description in the final technical specification.
- Validate all case-study metrics and define exactly how each metric is measured, against what baseline, and under what experimental conditions.
- Clarify whether posterior falsification always sets probability to zero or whether a calibrated Bayesian implementation is used in production.
- Define the operational meaning of low, medium, and high risk and map each class to explicit policy controls.

12.4 Recommended supporting presentation documents

DOCUMENT	PURPOSE	WHEN TO USE
Executive One-Pager	Problem, solution, differentiator, proof points, deployment concept.	Before or after the presentation.
Architecture Diagram	Agent, hardware, data, security, and governance layers.	Technical stakeholder discussions.
Case Study Sheet	One-page problem → method → outcome format for each vertical.	Customer discovery and demos.
Security & Governance Brief	Risk gates, provenance, deployment isolation, human approvals.	CTO, security, compliance reviews.
Demo Script	Campaign creation → hypothesis → approval → run → telemetry → update → audit.	Live product demonstration.
Evidence Pack	Supporting experiments, benchmarks, references, and validation artifacts.	Due diligence and scientific review.

End of HELIX Technical Master Dossier & Presentation Guide